

DETERMINING AN ISSUER PREFIX CONFIGURATION

Issuer Determine

- Issuer determination processing involves configuration for:
 - Prefix (on-us)
 - System Prefix (not-on-us)



On-us Prefixes

- On-us prefixes can also be called local prefixes. These are prefixes (BINs) used by Institutions in the BASE24-eps network.
- Also called “known” prefixes



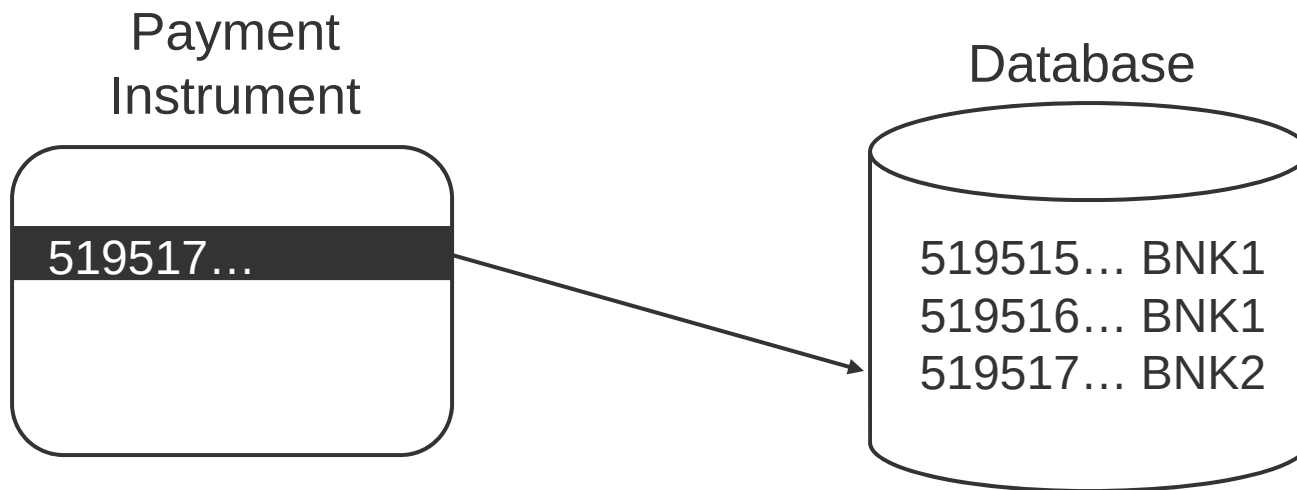
Not-on-us Prefixes

- Not-on-us prefixes are also called unknown prefixes.
- Typically, not enough is known about cards issued under these prefixes to allow the BASE24-eps network to authorize the transaction. Therefore, authorization is usually provided by an external processor.

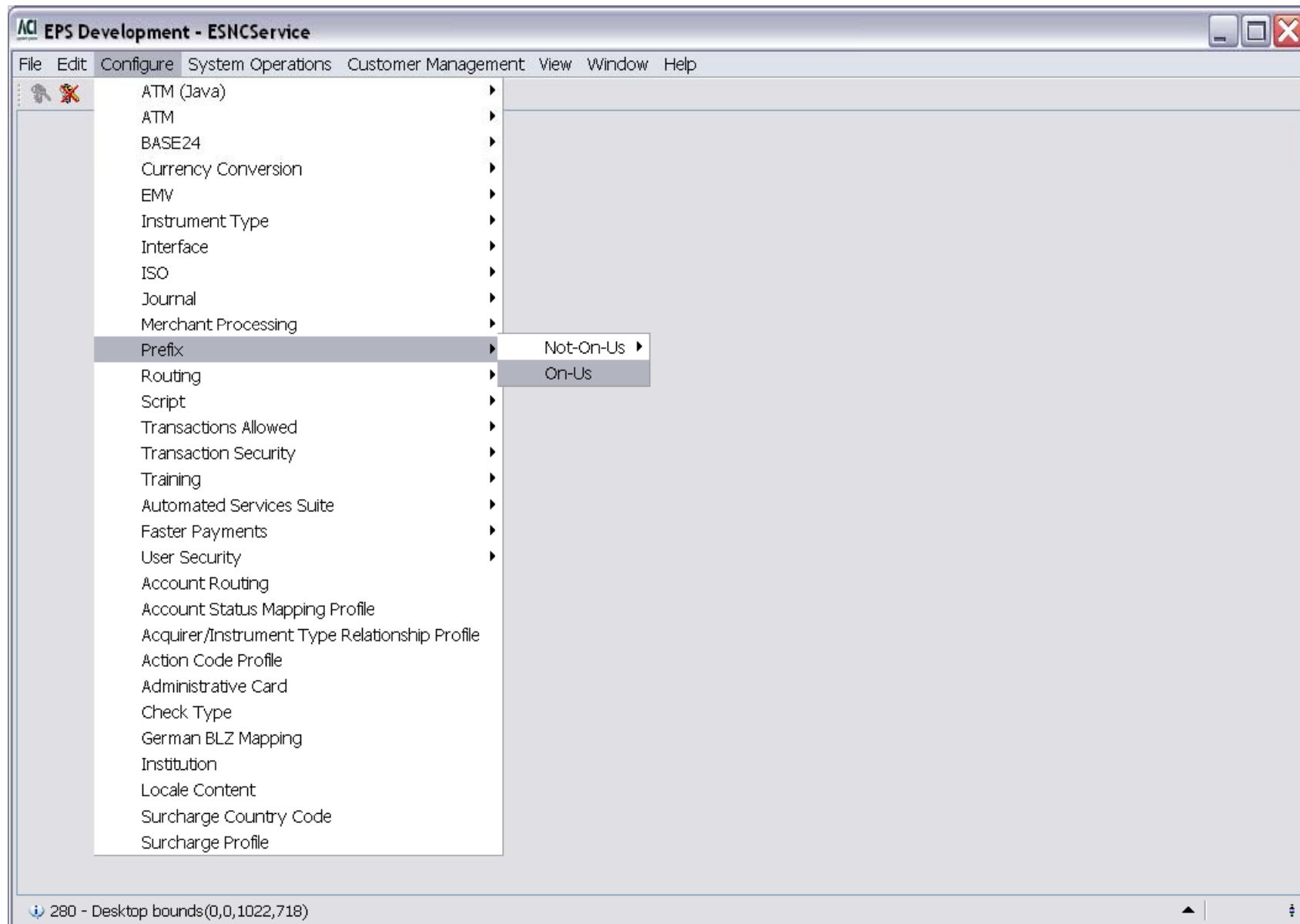


Determining an Issuer

- Determining an issuer involves:
 - retrieving an identifier (usually a prefix number) from the payment instrument (e.g., card)
 - matching the identifier retrieved from the payment instrument to a database entry



Prefix (on-us)



Prefix (on-us)

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File Edit Configure System Operations Customer Management View Window Help

ACI On-Us Prefix Configuration

Prefix PAN Length

Issuer ID

Card Verification Dynamic Card Verification EMV General Options EMV Auth Options CAVV

Processing Information Check Processing Issuer Information Card Track Information PIN Verification

☐ Member Number on Track

☐ Include Holds When Checking Limits, Usages and Balances

Expiration Date Check Option

Expiration Date Comparison Option

Fraud System Option

Maximum Allowed PIN Tries

Check Digit Algorithm Type

PIN Retry Limit Exceeded Action

PIN Try Reset Option

Route Type

☒ Acquirer and Issuer

☐ Issuer only

Default Preauthorization Hold

Time Minutes

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A match on a prefix for a given PAN length yields the Issuer ID and prefix related processing parameters.

Prefix (on-us)

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On-Us Prefix Configuration

Prefix PAN Length

Issuer ID

Card Verification Dynamic Card Verification EMV General Options EMV Auth Options CAVV

Processing Information Check Processing Issuer Information Card Track Information PIN Verification

☐ Include In Total Cash Advance/Withdrawal Usage for Checks Not Found on the Check Status File

☐ Include In Total Cash Advance/Withdrawal Usage for Checks Found on the Check Status File

☐ Check the Stop Payment File Before Sending Check Transactions to the Host

☐ Check the Check Status File Before Sending Check Transactions to the Host

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The Check Status File contains pre-approved/ pre-declined or registered checks. The Stop Payment file lists checks that are not honored.

Prefix (on-us)

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ACI On-Us Prefix Configuration

Prefix PAN Length

Issuer ID

Card Verification Dynamic Card Verification EMV General Options EMV Auth Options CAVV

Processing Information Check Processing Issuer Information Card Track Information PIN Verification

Instrument Type

Instrument Scheme

Member Length

Destination Routing Profile

Issuer Transactions Profile

Limit Profile

Issuer Surcharge Profile

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The Destination Routing Profile is used to determine authorization routing for requests. Destination Routing Profile configured in the Institution record is used for chargeback transactions and transactions with no PAN present.

Prefix (on-us)

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File Edit Configure System Operations Customer Management View Window Help

On-Us Prefix Configuration

Prefix: Select/Enter PAN Length: Select/Enter

Issuer ID:

Card Verification Dynamic Card Verification EMV General Options EMV Auth Options CAVV

Processing Information Check Processing Issuer Information Card Track Information PIN Verification

Track Processing Sequence

Primary: Track 2 Secondary: Track 1 Tertiary: Track 3

Track 1 Settings Track 2 Settings Track 3 Settings

Track Length

Maximum: 0 Minimum: 0

Card Verification: 3 Dynamic Card Verification: 0 Application Tran Counter: 0

Location on Track

Member Number: 0 Service Code: 0 Dynamic Card Verification: 0

PVV Key Index: 0 Expiration Date: 0 Application Tran Counter: 0

PVV: 0 Card Verification: 0 Unpredictable Number: 0

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Track Settings indicate the position of the respective data on the magnetic stripe of the card.

Prefix (on-us)

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On-Us Prefix Configuration

Prefix PAN Length

Issuer ID

Card Verification Dynamic Card Verification EMV General Options EMV Auth Options CAVV

Processing Information Check Processing Issuer Information Card Track Information PIN Verification

PIN Location

☐ No PIN Verification Number available

☐ PIN Verification Number on Cardholder File

☐ PIN Verification Number on Card Track

PIN Verification Type

PIN Verification Number Length

PIN Verification Profile

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PIN Verification specifies PIN Location, PIN Type, PIN Number Length, and PIN Verification Profile.

Prefix (on-us)

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On-Us Prefix Configuration

Prefix: Select/Enter PAN Length: Select/Enter

Issuer ID:

Processing Information | Check Processing | Issuer Information | Card Track Information | PIN Verification

Card Verification | Dynamic Card Verification | EMV General Options | EMV Auth Options | CAVV

Card Verification Type:

Card Verification Profile:

Effective Date: March 2011 Manual Date Check:

Card Verification Retry: ☐

Manual Card Verification Type: No CVD2 required

Manual Card Entry Option: Do not deny manually entered transactions

Card Verification Failure Action

Complete Track:

Uncertain Track:

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Card Verification specifies card verification checks and actions.

Prefix (on-us)

The screenshot shows a software window titled "EPS Development - ESNCServices" with a menu bar (File, Edit, Configure, System Operations, Customer Management, View, Window, Help) and a toolbar. Inside, there is a sub-window titled "On-Us Prefix Configuration" with its own toolbar. The configuration area includes fields for "Prefix" (set to "Select/Enter"), "PAN Length" (set to "Select/Enter"), and "Issuer ID". Below these are several tabs: "Processing Information", "Check Processing", "Issuer Information", "Card Track Information", "PIN Verification", "Card Verification", "Dynamic Card Verification" (which is selected), "EMV General Options", "EMV Auth Options", and "CAVV". The "Dynamic Card Verification" tab contains the following settings:

- "Dynamic Card Verification Type" is set to "Do not check Dynamic Verification Card Digits".
- "Dynamic Card Verification Profile" is an empty dropdown.
- "Pre-screen Dynamic Card Verification Check" is an unchecked checkbox.
- "Bad Dynamic Card Verification Action" is set to "Deny and return the card".

At the bottom of the main window, a status bar displays "280 - Desktop bounds(0,0,1022,718)".

Dynamic Card Verification specifies support for contactless chip transactions such as VISA Wave and MasterCard PayPass.

Prefix (on-us)

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File Edit Configure System Operations Customer Management View Window Help

On-Us Prefix Configuration

Prefix: Select/Enter PAN Length: Select/Enter

Issuer ID: [Dropdown]

Processing Information		Check Processing		Issuer Information		Card Track Information	
PIN Verification	Card Verification	Dynamic Card Verification	EMV General Options	EMV Auth Options	CAVV		
Primary EMV Issuer Application Data Format		None					
Alternate EMV Issuer Application Data Format		None					
Card Authentication Type		Do not perform CAM check					
EMV DKI Security		☐		EMV Profile		[Dropdown]	
Transport Key Locator		[Text Box]		EMV Script MAC Length		4	
EMV Destination Routing Profile		[Dropdown]		EMV Card Verification Results Profile		[Dropdown]	
EMV Terminal Verification Results Profile		[Dropdown]		EMV Reason Online Code Profile		[Dropdown]	
EMV Card Verification		Offline PIN Support					
EMV Check Type		Do not perform card verification		Effective Date		January 2005	
EMV Check Method		Use non-EMV card verification		EMV Service Code		000	

EMV specifies the EMV scheme and actions. This information is placed in the EMV Data TDE.

EMV Card Verification tab

The BASE24-eps EMV Issuer service extension module uses these fields to determine whether to perform card verification on EMV transactions.

Prefix (on-us)

ACI BASE24-eps Development - ESNCServices

File Edit Configure System Operations Customer Management View Window Help

On-Us Prefix Configuration

Prefix: Select/Enter PAN Length: Select/Enter

Issuer ID: [Dropdown]

Processing Information | Check Processing | Issuer Information | Card Track Information

PIN Verification | Card Verification | Dynamic Card Verification | EMV General Options | EMV Auth Options | CAVV

Primary EMV Issuer Application Data Format: None

Alternate EMV Issuer Application Data Format: None

Card Authentication Type: Do not perform CAM check

EMV DKI Security: ☐ EMV Profile: [Dropdown]

Transport Key Locator: [Text Box] EMV Script MAC Length: 4

EMV Destination Routing Profile: [Dropdown] EMV Card Verification Results Profile: [Dropdown]

EMV Terminal Verification Results Profile: [Dropdown] EMV Reason Online Code Profile: [Dropdown]

EMV Card Verification | Offline PIN Support

Bad PIN Tries Action: Do not perform EMV specific processing

PIN Management: Do not perform EMV specific processing

☐ Use PIN Support Date: July 2013

Offline PIN Support tab

This information is placed in the EMV Data TDE.

These fields control the actions BASE24-eps performs for the management of offline PINs. All of the fields in this section of the tab are carried in the Offline PIN Management (OPINMG) TDE.

Prefix (on-us)

ACI On-Us Prefix Configuration

Prefix: PAN Length:
Issuer ID:

Processing Information | **Check Processing** | **Issuer Information** | **Card Track Information**

Pre-screening Options

- ☐ Card Authentication Verification
- ☐ Application Transaction Counter Verification
- ☐ Card Verification Results
- ☐ Terminal Verification Results
- ☐ Reason Online Code Verification
- ☐ Fallback Check

Authorization Options

- ☐ Card Verification Results
- ☐ Terminal Verification Results
- ☐ Reason Online Code Verification
- ☐ Force Fallback to Over-the-Floor Indicator

Application Transaction Counter Verification:
Next Online Date Period:

Card Authentication Failure Action

Reliable: Unreliable:

Fallback Action: ATC Bad Disposition Action:

278 - Security Information - Last successful login: 2/19/12 4:47 PM

Configures options for prescreening and authorizing EMV transactions. This data is placed in the EMV data TDE except for Next Online Date Period field value which has it's own TDE.

Prefix (on-us)

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File Edit Configure System Operations Customer Management View Window Help

On-Us Prefix Configuration

Prefix PAN Length

Issuer ID

Processing Information Check Processing Issuer Information Card Track Information PIN Verification

Card Verification Dynamic Card Verification EMV General Options EMV Auth Options CAVV

Secure Internet Validation Check Type

CAVV Profile

Secure Internet Validation Pre-screening Check

CAVV Authentication Failure Action

CAVV Validation Check Failure Action

CAVV Not Present Action

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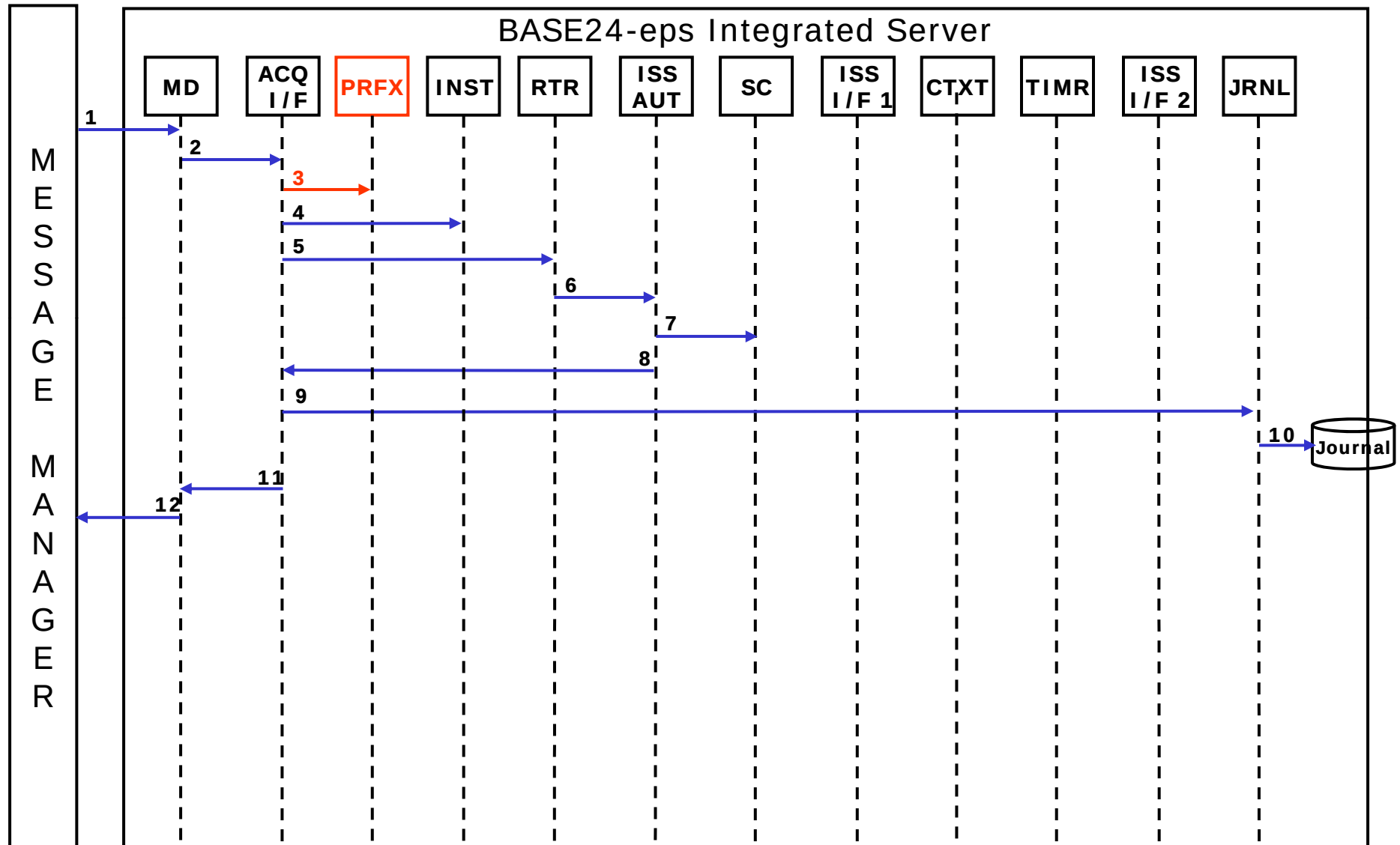
Cardholder Authentication Verification Value (CAVV). Secure Internet Validation - 3-D Secure Electronic Commerce Verification Service, Verified by Visa and Visa Payer Authentication.

Prefix (on-us)

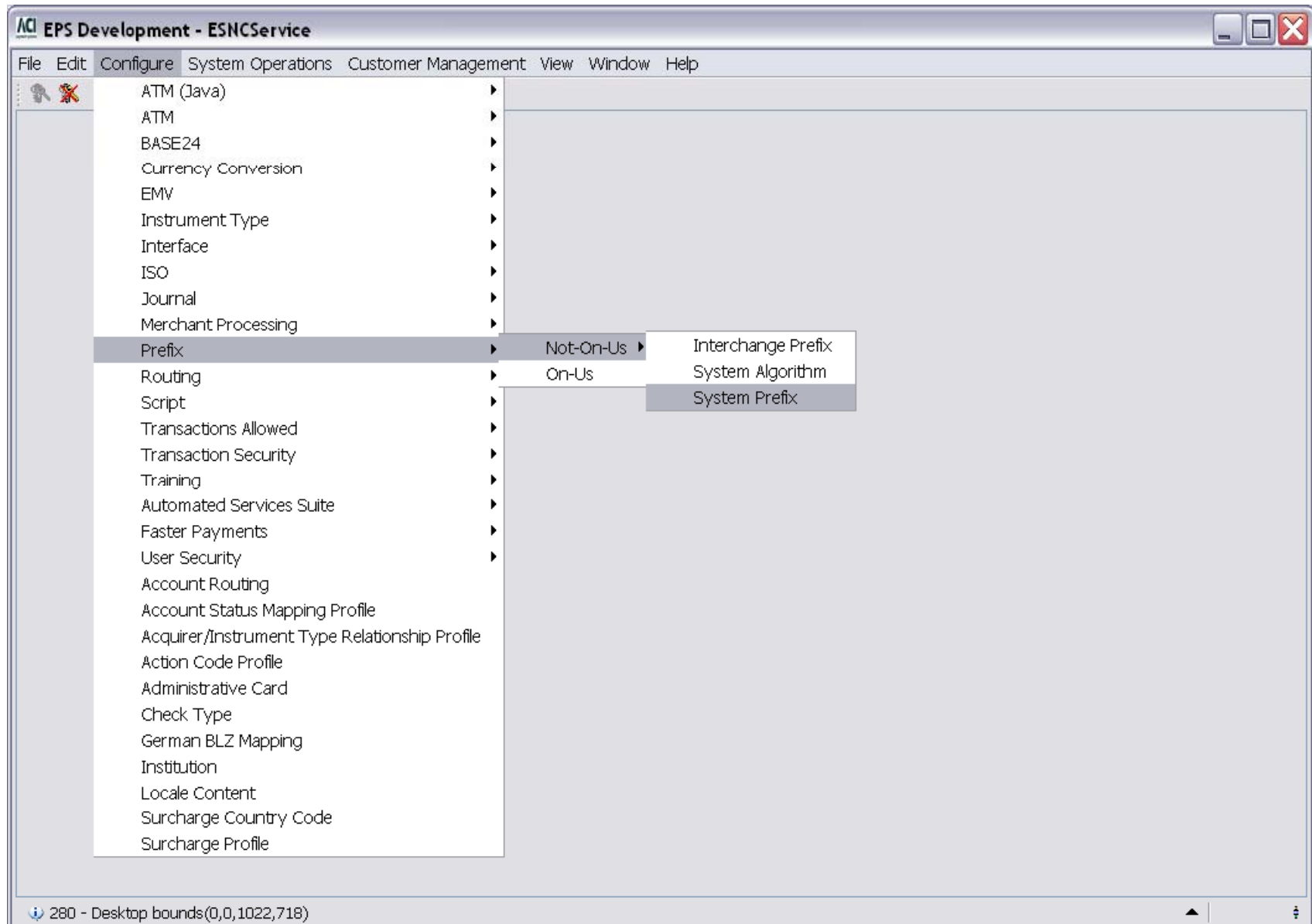
- Changes to the On-us Prefix configuration table requires rebuilding and reloading the respective OLTP table(s):
 - prefix_oltp
- OLTP tables can be rebuilt and loaded using Process Control or the OLTP Warmboot Management UI.



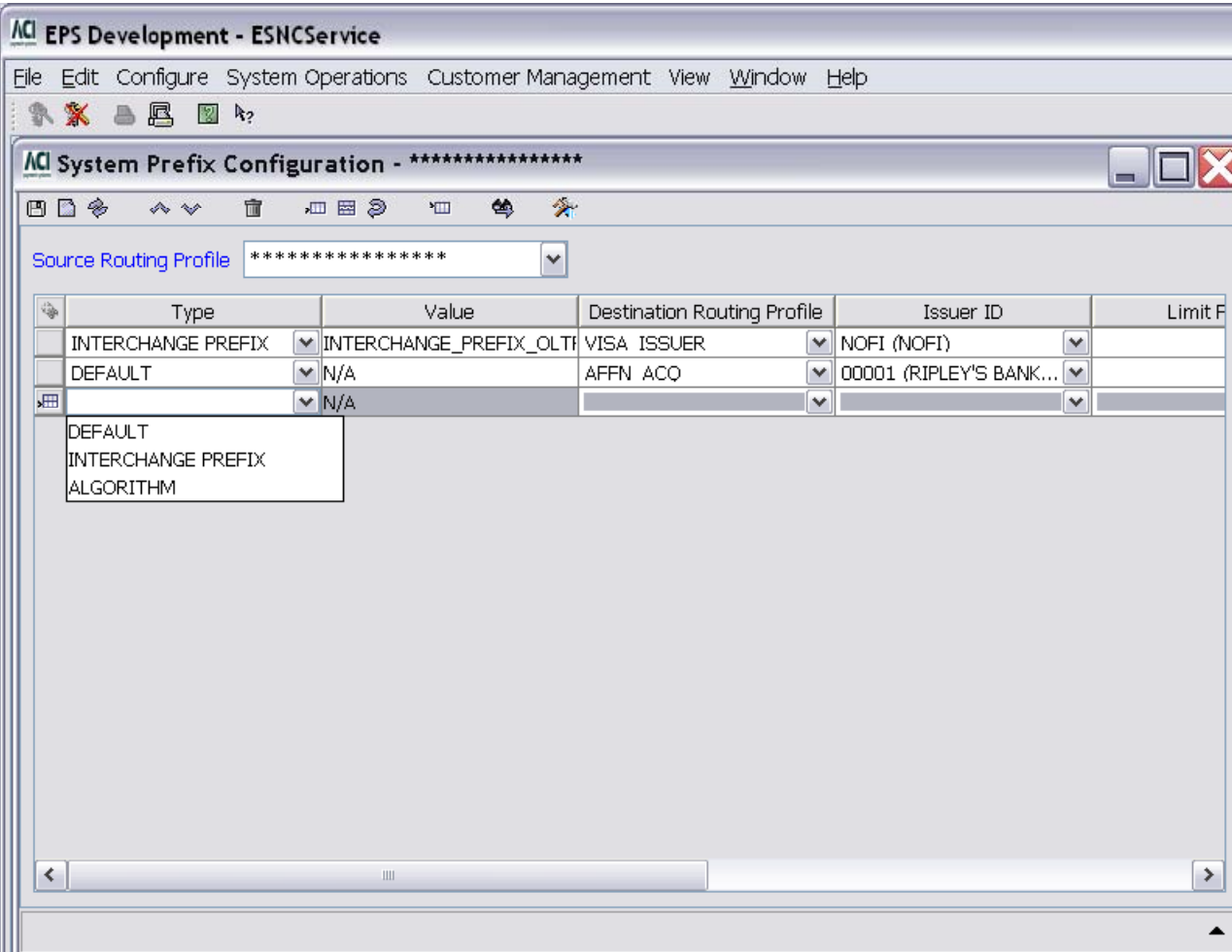
Component Interactions



Prefix (not-on-us)



Prefix (not-on-us)



Source Routing Profile *****

Type	Value	Destination Routing Profile	Issuer ID	Limit F
INTERCHANGE PREFIX	INTERCHANGE_PREFIX_OLTI	VISA ISSUER	NOFI (NOFI)	
DEFAULT	N/A	AFFN ACO	00001 (RIPLEY'S BANK...)	
DEFAULT	N/A			

Additional rows can be added. Options include performing an Interchange Prefix search, using an algorithm, or utilizing a default method.

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Prefix (not-on-us)

Source Routing Profile *****

Issuer ID	Limit Profile	Issuer Surcharge Profile	Instrument Type	Route Type
FI)			VH - Visa Plus	Issuer only
PLEY'S BANK...			AF - Armed Forces Fi...	Acquirer & Issuer

Route type indicates whether or not the source of the transaction will be considered in the routing decision.

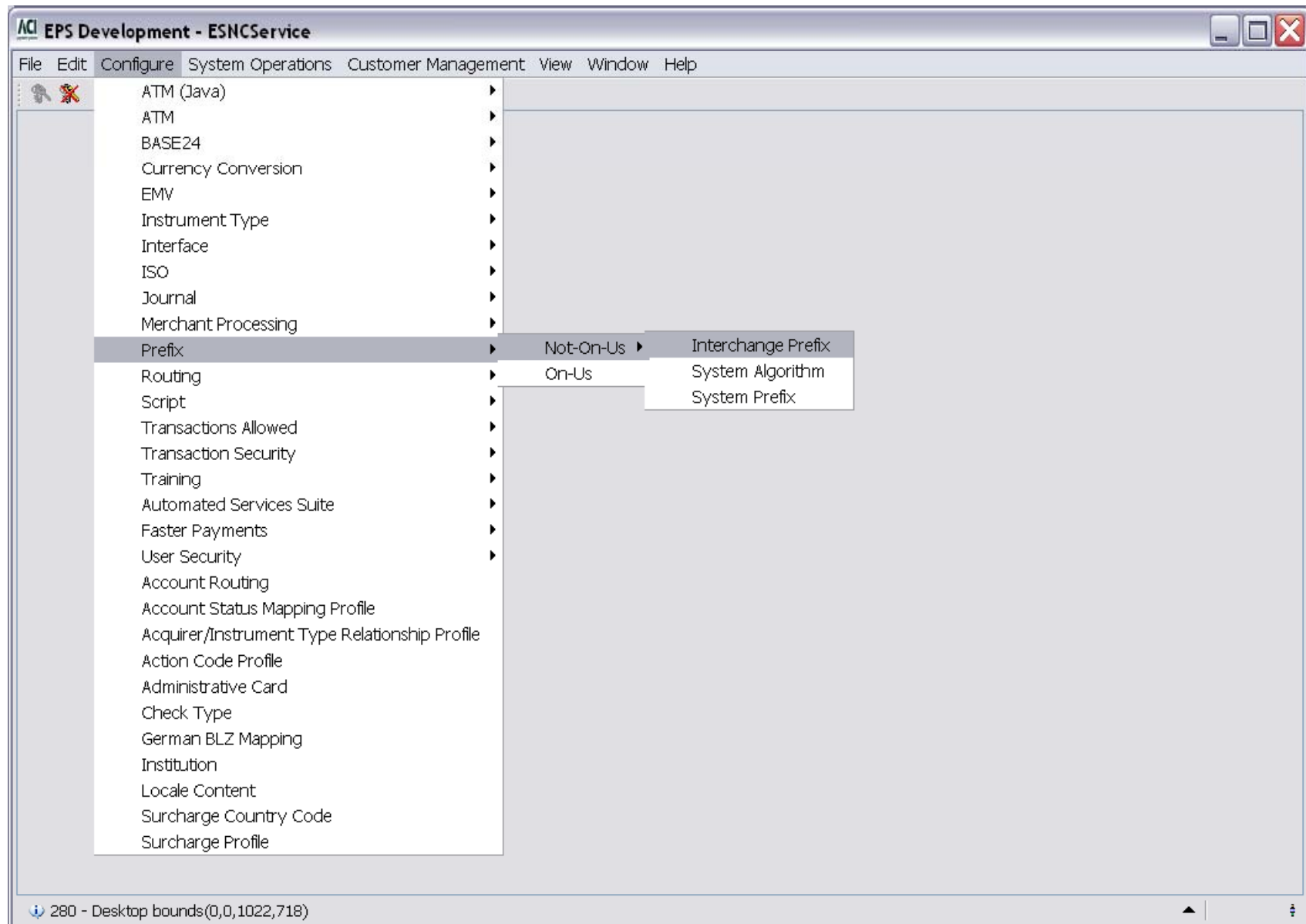
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Prefix (not-on-us)

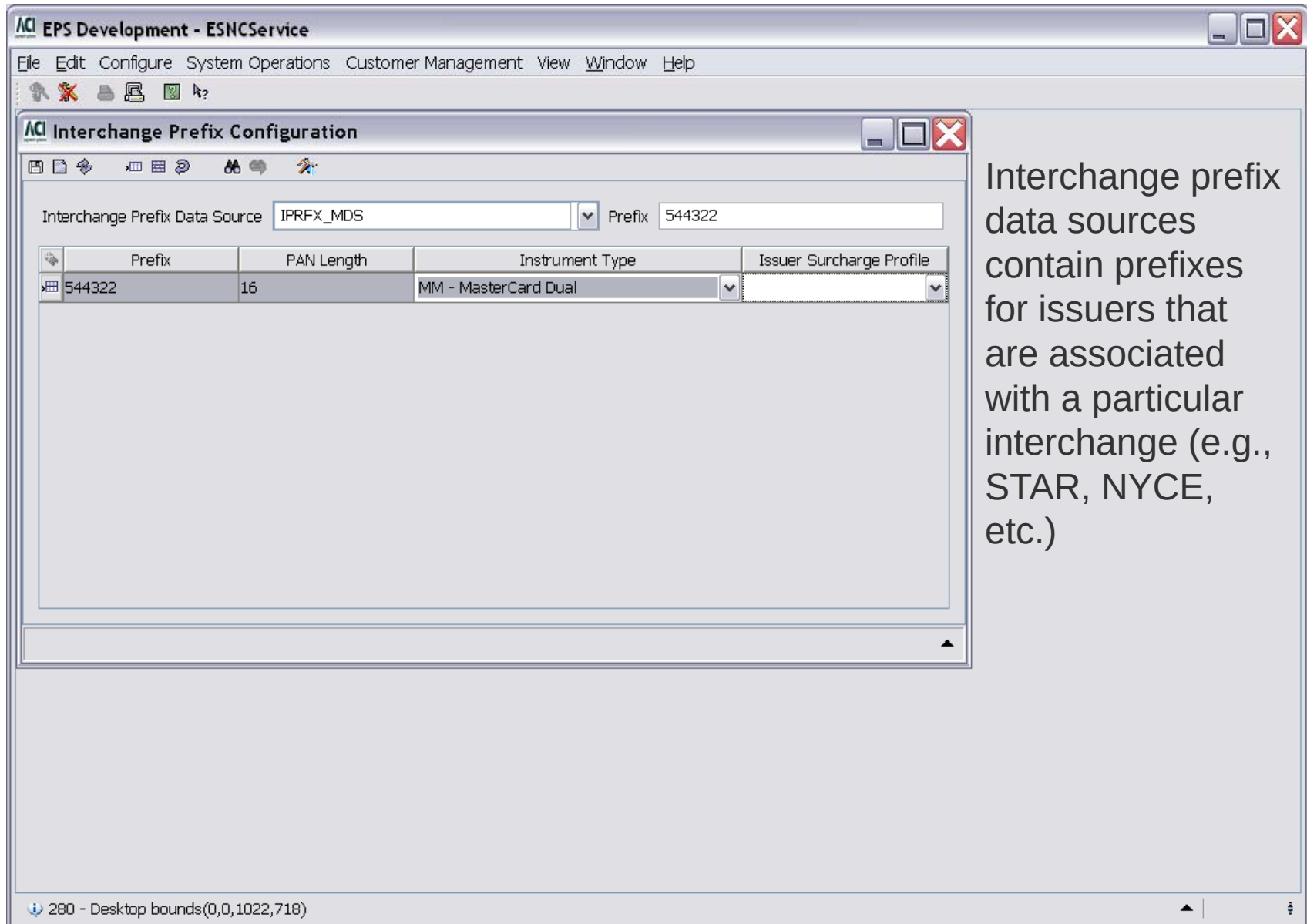
- Based on the Source Route Profile value provided by the Acquirer Interface, the System Prefix configuration is searched for the best match. Once found, the entry provides:
 - Destination Routing Profile name: used by Router to find routing options
 - Issuer ID: Used to provide the Issuer journal set
 - Limit Profile: Could be used in a pre-screen script
 - Issuer Surcharge: surcharging alliances
 - Instrument Type: could be used in reports
 - Route Type: determines Route Code; Issuer only or Acquirer and Issuer



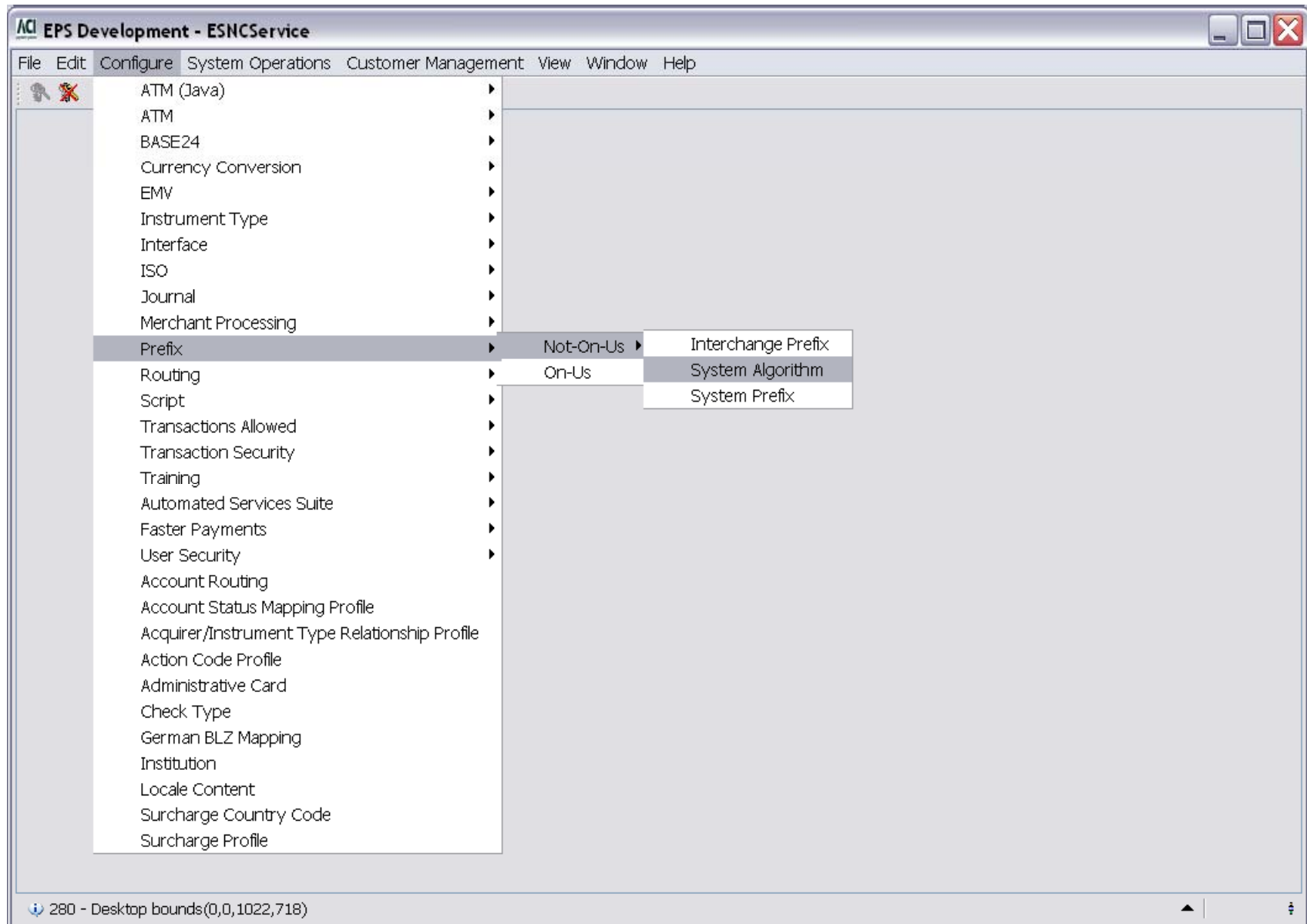
Prefix (not-on-us)



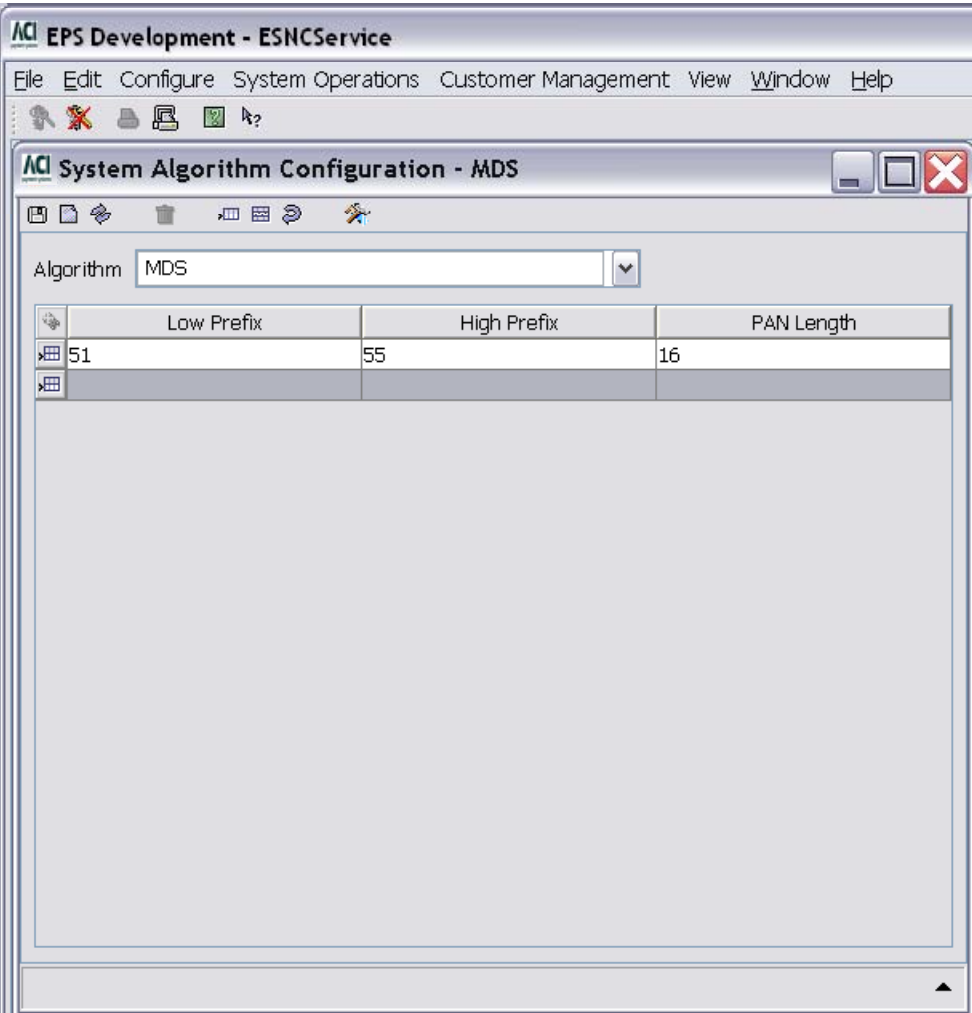
Prefix (not-on-us)



Prefix (not-on-us)



Prefix (not-on-us)



Algorithm: MDS

Low Prefix	High Prefix	PAN Length
51	55	16

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Algorithms can be applied against a PAN to determine the issuer. Standard algorithms can be defined for Visa, MasterCard (INAS or BNET), Discover, American Express and Diners Club. Any additional user defined algorithm can be created and used to determine the issuer.

Prefix (not-on-us)

- Additions to the interchange prefix data source table(s) requires rebuilding and reloading the respective OLTP table(s). For example:
 - iprfx_mds_oltp
 - iprfx_visa_oltp
 - etc.
- OLTP tables can be rebuilt and loaded using Process Control or the OLTP Warmboot Management UI.

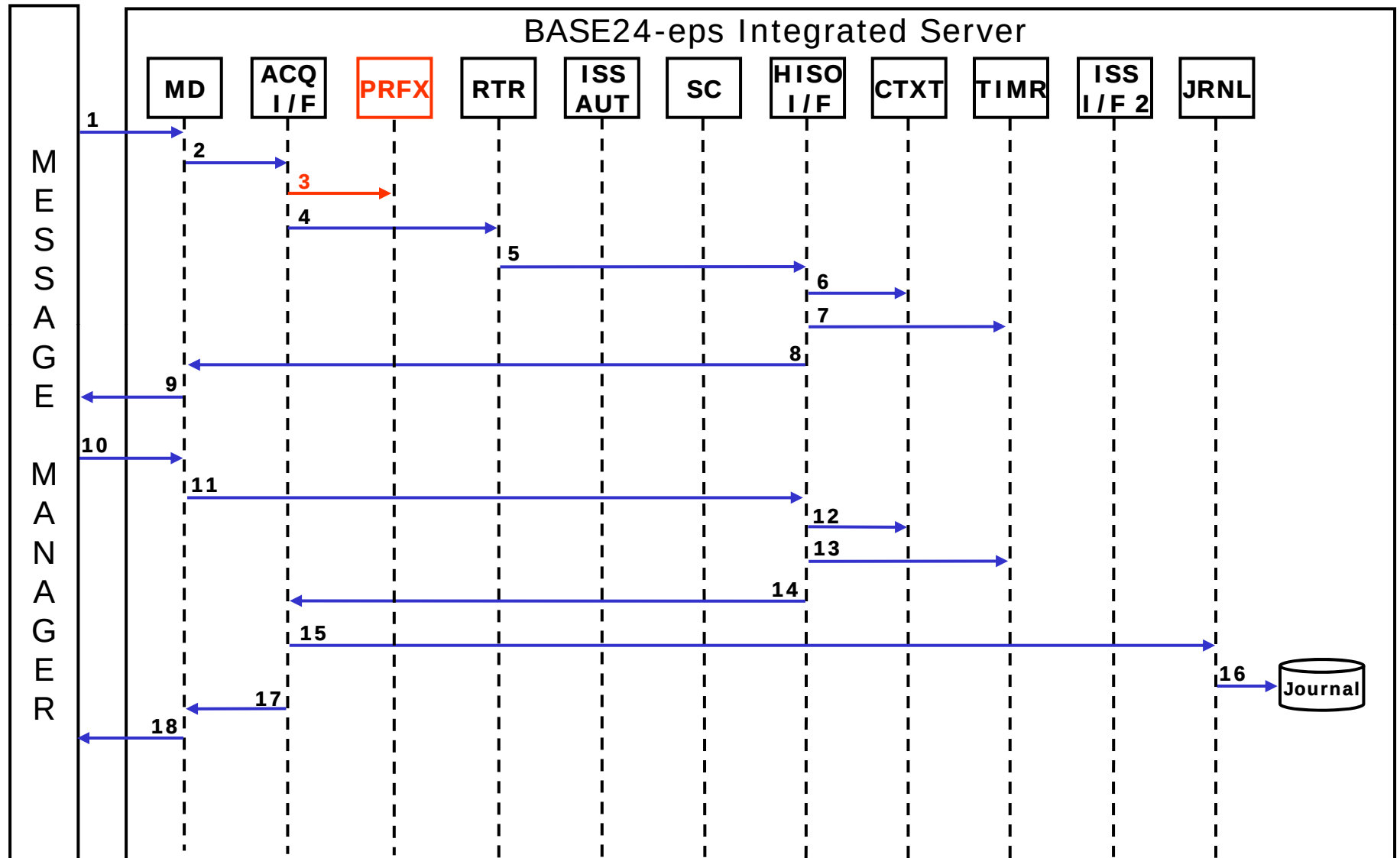


Prefix (not-on-us)

- Changes to the not-on-us Prefix configuration System Algorithm, and/or System Prefix table(s) requires rebuilding and reloading the respective OLTP table(s):
 - system_algorithm_config_oltp
 - system_prefix_oltp
- OLTP tables can be rebuilt and loaded using Process Control or the OLTP Warmboot Management UI.



Component Interactions



External Authorization (Not-On-Us Prefix) – Request/Response

Lab Exercise

BASE24-eps Prefix Configuration Lab

Lab 6





MEETS THE CHALLENGE OF CHANGE